

**Gemini
Near
Infra-Red
Imager
and
Wavefront
Sensor**



Near IR Imager (NIRI) — Hardware Notes

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This report contains a few notes that may be useful for configuring NIRI hardware.

1 Introduction

This report contains a few notes that are useful in configuring the NIRI hardware.

1.1 Purpose of the document

This report contains a few informal notes which may be useful to anyone who has to configure or maintain NIRI hardware.

The intended audience of this document is:

- the maintainers of the NIRI software and hardware.

2 The Motor Driver Modules

The following table lists the current settings for all motor driver modules at the time that NIRI was initially shipped. Note that if necessary, the current may be raised to overcome sticking in the mechanisms; however, this is frequently a sign of a mechanical problem which should be resolved. The currents listed have been found to be sufficient for reliable operation at room temperature and at cryogenic temperatures. Excessive current settings should be avoided because of the added heat load to the system and the possibility of significant thermal gradients across mechanisms.

TABLE 1. Motor Driver Modules: Current settings

		Column 1	Column 2	Column 3	Column 4	Column 5
Row 1	Mechanism	Pupil Mask / Filter Wheel 3	Filter Wheel 1	Filter Wheel 2	Steering Mirror 1	Steering Mirror 2
	Current Setting ^a	2	2	2	2	3.75
Row 2	Mechanism	Empty	Science Detector Focus	Pupil Viewer	Gimbal x	Focal Plane Mask
	Current Setting ^a		3.75	1	1.25	2.5
Row 3	Mechanism	Gimbal y	Beam Splitter	WFS Focus	WFS Filter	Window Cover
	Current Setting ^a	1	1.25	1	2	2

- a. The current settings on the Phytron driver modules go from slightly less than 0 to slightly more than 4, in unspecified units. The actual range of current values is 0.6 A to 2.6 A.